

Report ID: REP-20260829184452-ff372f

Compliance: COMPLIANT (Strong IPsec Security Posture)

Target File: final_ipsec.pcap

Timestamp: Aug 29, 2026, 18:44 UTC

1. EXECUTIVE SECURITY SUMMARY & POSTURE GRADE

GRADE A+

Normalized Risk Index: 10 / 100 (LOW RISK)

Status: Verified by TITAN Engine

2. CRYPTOGRAPHIC INTEGRITY & POST-QUANTUM (PQC) READINESS

- Payload Encryption: Enforced (ESP Protocol 50)

- Authentication Header: None (Pure ESP Tunnel)

- Active SPIs: 0xc743e05b, 0xcb68f974
- Post-Quantum Readiness: 85% (QUANTUM-RESISTANT (CNSA 2.0 Symmetric Key))

- IKE PSK Exposure Risk: LOW (No IKEv1 Aggressive Mode Hash Exposure)

- CNSA 2.0 Compliance: COMPLIANT

3. TRAFFIC TELEMETRY & PROTOCOL DISTRIBUTION MATRIX

Protocol Layer / Name	Packet Count	Traffic Share (%)	Security Classification
ESP Payload Encryption (Protocol 50)	60	100.0%	Encrypted (Secure)
Authentication Header AH (Protocol 51)	0	0.0%	Cleartext / Control
IKE / NAT-T (UDP Port 500 / 4500)	0	0.0%	Cleartext / Control
TCP Transport Streams	0	0.0%	Cleartext / Control
UDP Datagrams (Non-VPN)	0	0.0%	Cleartext / Control
DNS Resolution Queries	0	0.0%	Cleartext / Control
ICMP Diagnostic Messages	0	0.0%	Cleartext / Control
TOTAL CAPTURED TRAFFIC	60	100.0%	Full Network Stream

4. ENCRYPTED TRAFFIC ANALYSIS (ETA) & APPLICATION FINGERPRINTING

- Inferred Application: Interactive Shell / VoIP / Command Stream

- Traffic Pattern: Low Latency / Fixed Size Frames

Behavioral Heuristic: Continuous stream of small, fixed-size encrypted frames indicative of interactive management sessions or voice-over-IP.
- ETA Confidence Score: 94.5%

- Burstiness Index: 0.0

5. MITRE ATT&CK ENTERPRISE FRAMEWORK MAPPING

Technique ID	Technique Name	Tactic / Domain	Severity	Audit Finding
M1037	Filter Network Traffic	Defensive Mitigation	INFO	100% of traffic is encapsulated

6. DEEP SECURITY AUDIT FINDINGS

- [1] ESP Payload Encryption Verified: 60 encrypted frames (100.0% of capture).
- [2] Full Encapsulation: 100% of captured traffic is encapsulated within secure IPsec tunnel.
- [3] Established Tunnel Trace: IKE initial negotiation phase was completed prior to packet capture.
- [4] Zero Cleartext Leakage: No plaintext transport protocols were observed bypassing the VPN tunnel.
- [5] MTU Headroom Optimal: Average packet length is 170 bytes (Max: 170 bytes), avoiding fragmentation.
- [6] Anti-Replay Protection: Sequence tracking verified across 2 active Security Associations (SAs).

7. ACTIONABLE HARDENING ROADMAP & SIEM CONFIGURATION

Step 1: Maintain current cryptographic parameters and schedule regular re-keying intervals (e.g. every 8-24 hours).